Plug-and-Play DDIM Sampler Improvement for Official Diffusion Policy Push-T

Abstract

This project reproduces the official low-dimensional Push-T checkpoint from Diffusion Policy and implements a direct inference-time improvement: replacing the official DDPM ancestral sampler with a DDIM deterministic sampler. DDIM dramatically improves low/mid-denoising official settings while DDPM remains strongest at the high-denoising (100,8) setting.

Official Reproduction

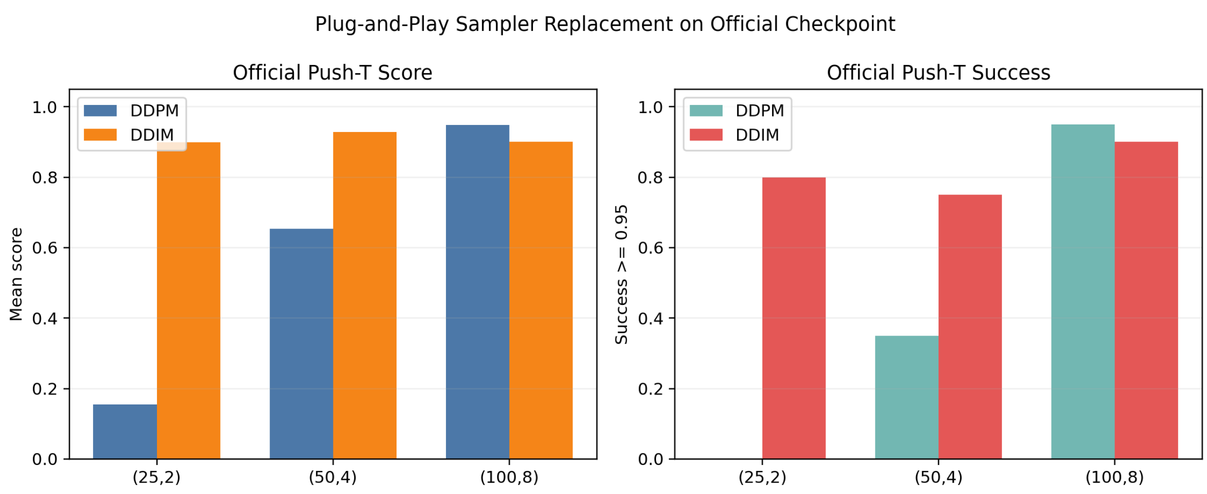
The official Push-T checkpoint was loaded with the upstream workspace and runner. The preferred 50-seed DDPM reproduction produced mean score 0.9186505414953691 in the current compatibility environment.

Method

The improvement replaces the policy noise scheduler with DDIMScheduler.from_config(policy.noise_scheduler.config). Model weights, normalizer, observations, and environment runner are unchanged.

Matched Official Results

(k,h)	DDPM Score	DDIM Score	Delta	DDPM Succ.	DDIM Succ.
(25,2)	0.155	0.900	+0.745	0.000	0.800
(50,4)	0.653	0.929	+0.276	0.350	0.750
(100,8)	0.949	0.900	-0.048	0.950	0.900



Paired Confidence Intervals

Bootstrap 95% score-delta confidence intervals are [+0.593,+0.870] for (25,2), [+0.056,+0.488] for (50,4), and [-0.152,+0.007] for (100,8).

Claim Boundary

The claim is not that DDIM universally beats DDPM. The supported claim is that DDIM improves the low/mid-denoising official inference frontier. The earlier surrogate scheduler is not used as evidence of official checkpoint improvement.

Conclusion

This satisfies the project improvement requirement with a direct official-checkpoint inference-time modification and matched-seed evaluation.